Law-Tech: An AI-Powered Indian Legal Assistant Using Retrieval-Augmented Generation (RAG)

Project Registration

1. Nilesh Sanyasi | vu2f2324013@pvppcoe.ac.in | VU2F2324013
2. Sairaj Gawade | vu2f2324008@pvppcoe.ac.in | VU2F2324008
3. Maitreya Channe | vu2f2324015@pvppcoe.ac.in | VU2F2324015
4. Sarthak Gavas | vu2f2324007@pvppcoe.ac.in | VU2F2324007

1. Problem Definition

- Indian laws, bare acts, and court judgments are written in complex legal language, making them largely inaccessible to ordinary citizens.
- Legal information is scattered across disconnected sources — statute portals, judgment databases, blogs — with no single reliable point of access.
- Professional legal consultation is expensive and out of reach for low-income citizens, while manual legal research is time-consuming even for those who can afford it.
- General-purpose AI chatbots (e.g. ChatGPT, Gemini) answer legal questions fluently but without reliable citations, and are prone to hallucinating incorrect legal provisions.
- There is no widely accessible system that combines accurate legal question-answering, citation to source acts/judgments, document analysis, and draft generation in one place.

Root Cause: Existing tools either provide legal information without verification (blogs, generic chatbots) or require paid human expertise (lawyers), with nothing in between that is both accurate and accessible. The missing piece is an AI system that retrieves from verified legal sources before answering, so that responses are grounded rather than guessed — an approach generic LLMs do not take by default.

2. Introduction

2.1 The Legal Accessibility Gap

India's legal framework — bare acts, amendments, and case law from the Supreme Court and High Courts — is vast, technical, and written for legal professionals rather than the public. Citizens seeking to understand their rights or obligations must either navigate this material themselves, search across scattered and often unofficial sources, or pay for a lawyer's time. This creates a persistent gap between the law as written and the law as understood, particularly for low-income citizens, students, and small businesses who cannot easily afford repeated legal consultations.

2.2 Existing Digital Tools and Their Limitation

Generic AI chatbots can discuss legal topics in plain language but generate answers from learned patterns rather than verified source documents, making them prone to hallucinated sections, incorrect citations, or outdated law. Legal information portals and blogs provide static content without interactive question-answering. Search across bare acts and judgments individually still requires the user to already know which act or case to look for. None of these combine accurate retrieval from verified legal sources with a conversational interface and traceable citations.

2.3 Law-Tech — The Proposed Solution

This project proposes Law-Tech, an AI-powered Indian legal assistant built on Retrieval-Augmented Generation (RAG). Rather than generating answers purely from a language model's learned knowledge, Law-Tech retrieves relevant sections of bare acts and court judgments from a verified legal database before generating a response, and returns every answer together with its source citation. The system additionally supports OCR-based analysis of uploaded legal documents, generation of standard legal drafts, and multilingual/voice interaction, combining these into one integrated assistant rather than separate point tools.

3. Literature Review

Paper	Year	Contribution	Limitation / Gap
Law-Tech: AI-Powered Legal Assistant Using RAG, Int. J. of AI & Law	2026	Presents a RAG-based architecture for legal question-answering with citation grounding.	General framework; does not integrate OCR document analysis, draft generation, and multilingual voice support in one platform.
Aalap: AI Assistant for Legal and Paralegal Functions in India, AAAI	2024	AI assistant fine-tuned for Indian paralegal tasks such as drafting and summarization.	Focused on paralegal drafting tasks; lacks a retrieval-grounded citation mechanism for general public queries.
Legal Assist AI Using Transformer Models for Indian Judicial Systems, IEEE Access	2025	Transformer-based model for analyzing Indian judicial text.	Model-centric study; no end-to-end citizen-facing chatbot, no document upload/OCR pipeline.
Domain-Partitioned Hybrid RAG for Legal Reasoning in Low-Resource Settings, ACL Findings	2026	Hybrid RAG architecture that partitions retrieval by legal domain (criminal, civil, corporate) to improve reasoning accuracy.	Addresses retrieval architecture only; no discussion of user-facing features such as document generation or multilingual support.
AI Legal Assistant Using RAG and Large Language Models: A Survey, arXiv	2026	Surveys RAG-based legal assistant approaches across jurisdictions.	Survey-level; does not propose or evaluate a specific integrated system for Indian law.

Across the surveyed work, RAG-based grounding for legal question-answering is an active research direction, but no reviewed system combines citation-grounded Q&A, OCR-based document analysis, legal draft generation, and multilingual/voice access into a single platform targeted at Indian citizens, students, and legal professionals — the gap Law-Tech is designed to close.

4. Motivation

- Accuracy Requires Grounding: A legal answer without a verifiable source is not trustworthy; AI assistance for law is only useful if every answer can be traced back to the actual bare act or judgment it is based on.
- Legal Awareness Is a Public Good: Understanding one's legal rights should not depend on the ability to pay for a consultation — an accessible, always-available assistant can meaningfully close this gap for students, low-income citizens, and small businesses.
- Generic AI Is Not Enough: General-purpose chatbots are fluent but not reliable for law, since they are not grounded in verified legal sources and can hallucinate provisions or misattribute judgments.

- Technology Has Matured Enough to Help: Advances in retrieval-augmented generation, vector search, OCR, and multilingual NLP now make it feasible to build a citation-grounded legal assistant that is both accurate and accessible at low cost.

Law-Tech is motivated by the gap between what citizens need — accurate, citation-backed, understandable legal information — and what currently exists, which is either unreliable generic AI or expensive human expertise, with no accessible middle ground.

5. Objectives

- G1 — Design and implement a Retrieval-Augmented Generation pipeline that retrieves relevant bare-act sections and judgments before generating an answer.
- G2 — Build a legal chatbot module that answers user queries in plain language with source citations for every response.
- G3 — Develop a document analyzer that uses OCR to extract and analyze text from uploaded legal documents.
- G4 — Build a legal document generator that produces standard drafts such as affidavits, agreements, and notices from templates.
- G5 — Support voice input and multiple Indian languages to widen accessibility.
- O1 — Achieve high retrieval accuracy, ensuring the correct bare-act sections or judgments are retrieved for a given query.
- O2 — Validate citation accuracy, confirming that cited sources are genuine and correctly referenced.
- O3 — Measure OCR extraction accuracy on uploaded legal documents of varying quality.
- O4 — Integrate the chatbot, document analyzer, judgment search, and document generator into a single working assistant and demonstrate it end-to-end.

6. Methodology Used

6.1 System Architecture

Law-Tech is composed of a frontend (Flutter for mobile, React.js for web) through which users submit text or voice queries; a FastAPI backend that embeds the query and performs a vector similarity search; a vector database (FAISS/ChromaDB) holding embeddings of official bare acts and judgments; and a large language model (GPT/Llama 3) that generates the final answer from the retrieved documents. PostgreSQL stores user data, sessions, and conversation history, while all legal source text is vectorized and stored separately for fast semantic retrieval.

6.2 Retrieval-Augmented Generation (RAG) Pipeline

A user query is first converted into a vector embedding. This embedding is compared against the vector database of bare acts and judgments to retrieve the most semantically similar legal documents. These retrieved passages are then passed to the LLM along with the original query, so that the generated answer is grounded in actual legal text rather than the model's unguided output. The final response is returned together with citations identifying the specific act, section, or judgment it was drawn from.

6.3 Document Analysis via OCR

Uploaded legal documents (PDFs or images) are processed using Tesseract OCR to extract text, which is then passed through the AI pipeline to identify and summarize key clauses, enabling users to understand uploaded contracts or notices without manual reading.

6.4 Legal Document Generation

Standard legal drafts — such as affidavits, agreements, and notices — are produced using a template-based generation system, populated with user-provided details and reviewed for structural correctness before being returned to the user.

7. Development Plan

Phase	Timeline	Deliverable	Success Criterion
1	Months 1–2	Requirement analysis, dataset collection of bare acts and judgments, finalized system architecture.	Approved architecture diagram covering frontend, backend, vector DB, and LLM layer.
2	Months 2–3	Legal database and vector embedding pipeline — bare acts and judgments indexed in FAISS/ChromaDB.	Vector search returns relevant documents for a benchmark set of test queries.
3	Months 3–4	RAG-based AI chat module — FastAPI service integrating retrieval and LLM generation with citations.	Chatbot returns accurate, cited answers for common legal queries in evaluation testing.
4	Months 4–5	OCR document analyzer and legal document generator modules.	OCR correctly extracts text from sample documents; generator produces valid draft documents from templates.
5	Month 6	Voice/multilingual support, full system integration, testing, documentation, and final demonstration.	All modules — chatbot, document analyzer, judgment search, document generator — operate as one integrated pipeline; report and demo completed.

Software: Python 3.11+ (FastAPI, LangChain), FAISS/ChromaDB, PostgreSQL, React.js (web dashboard), Flutter (mobile application), Tesseract OCR, OpenAI Whisper, GPT-5.5/Llama 3, Git/GitHub.

Hardware: Development machines with a minimum of an Intel i5/Ryzen 5 processor, 8GB RAM; a cloud or on-premises server with GPU access for hosting the vector database and LLM inference; Android/iOS devices for mobile-app testing.

8. Conclusion

Law-Tech addresses a concrete and pressing gap in access to legal information: the absence of a single platform that combines accurate, citation-grounded legal question-answering with document analysis and draft generation. Rather than relying on a generic chatbot's unguided output, the project grounds every answer in retrieval from verified bare acts and judgments, directly addressing the hallucination and citation problems that limit existing AI legal tools.

The platform's contribution is threefold: a Retrieval-Augmented Generation pipeline that grounds legal answers in verified sources, an OCR-based document analyzer for uploaded legal documents, and a template-based legal document generator — unified under one multilingual, voice-accessible interface. Future work will extend the platform with court case tracking and hearing reminders, lawyer appointment booking, a full multilingual voice assistant, offline access, state-specific law modules, and predictive legal analytics using historical case data.

9. References

[1] "Law-Tech: AI-Powered Legal Assistant Using Retrieval-Augmented Generation," International Journal of AI & Law, 2026.

[2] "Aalap: AI Assistant for Legal and Paralegal Functions in India," Proceedings of AAAI, 2024.

[3] "Legal Assist AI Using Transformer Models for Indian Judicial Systems," IEEE Access, 2025.

[4] "Domain-Partitioned Hybrid RAG for Legal Reasoning in Low-Resource Settings," ACL Findings, 2026.

[5] "AI Legal Assistant Using RAG and Large Language Models: A Survey," arXiv:2026.04521, 2026.